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TES-P-119.09

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BUSBARS AND SUPPORTING STRUCTURES

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TRANSMISSION ENGINEERING STANDARD

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1.0 SCOPE

This standard covers the criteria for design of busbars and their supporting structures for outdoor and indoor air insulated substations in Saudi Electricity Company (SEC) transmission system. Civil engineering requirements associated with these are dealt with in TES-P-119.18 and TES-P-119.19.

2.0 PURPOSE

The typical outdoor open-type substation arrangement consists essentially of open-bus construction using rigid bus design. The transmission line is terminated on take-off structures/gantry structures. From here the line drops into the bay and is connected to the substation equipment through rigid buses which are arranged to run along the length of the substation.

Indoor open-type stations comprise rigid bus running along the individual bays and connecting to the substation equipment installed in each bay, based on the required configuration. Incoming and/or outgoing feeders connecting to the outdoor equipment/structures would be via rigid bus or suitably rated cables.

Such an arrangement requires strain and rigid conductors, structures to support the transmission line conductors, support structures for disconnect switches, current transformers, CCVTs, lightning arresters etc., as well as structures for the strain and rigid buses in the station.

The purpose of this standard is to provide guidelines for satisfactory design of substation rigid-bus structure involving the electrical, mechanical and structural cosiderations. The steps involved in design of rigid bus at substation are shown in Figure 09-01.

3.0 AMPACITY

3.1 Current Rating of Bus

The ampacity (steady state current carrying capacity) of the bus-conductor is usually determined by either the electrical system requirements or the ampacity of the connected equipment. In both cases, the emergency and continuous overload ratings of the equipment/system shall be taken into consideration to arrive at the required current rating of the conductor.

3.2 Heat balance

The following factors are to be taken into account for determining the ampacity of the bus-conductor:

a. For outdoor substations, heat input due to I^2R and solar radiation equals the heat output due to convection, conduction and radiation.

For indoor substations, heat input due to I^2R equals the heat output due to convection, conduction and radiation.

b. Wind speed (For calculation of ampacity, conditions which are least advantageous for convection of heat must be considered, thus it is assumed that there is only 42 m/sec wind)

c. Full solar radiation (For SEC system at latitude 26°N, the value is 1.0kW/m²).

d. Emisivity factor (0.5 for outdoor and 0.35 for indoor).

e. For aluminum conductor conductivity 53% IACS (International Annealed Copper Standard) at 20°C.

f. Ambient temperature

Outdoor: monthly average of the hottest month as specified in 01-TMSS-01
Indoor: as specified in 01-TMSS-01 for substations with and without air-conditioning, as the case may be.

g. Maximum conductor temperature 85°C.

h. Altitude above mean sea level as specified in 01-TMSS-01.

Ampacity shall be based on the above factors and shall be calculated as per IEEE Standard 605 for Design of Substation Rigid Bus Structures.

3.3 Fault conditions

The conductor ampacity is limited by the conductor's maximum operating temperature.

3.3.1 Aluminum alloy and copper conductors may be operated continuously at 90ºC without appreciable loss of strength. Since the heat in the conductor may be conducted to attached equipment, conductor should not be operated at temperature high enough to damage the connected equipment.

3.3.2 A conductor's temperature will rapidly rise under fault conditions. The maximum allowable fault current can be calculated by:

For aluminum conductor(40% to 65% International Annealed Copper Standard (IACS) conductivity.

$$I = C \times 10^6 \times A \sqrt{\frac{1}{t}\log_{10}\left(\frac{T_f - 20 + \left(15150/G\right)}{T_i - 20 + \left(15150/G\right)}\right)} \qquad \text{(Eq.09.1)}$$

Where

I	=	maximum allowable root mean square (rms) value of fault current, Amperes
A	=	conductor crossectional area , mm^2
C	=	2.232×10^{-4} mm^2
G	=	conductivity in percent International Annealed Copper Standard
t	=	duration of fault in seconds
T_f	=	allowable final conductor temperature, ºC
T_i	=	conductor temperature at fault initiation, ºC

For copper conductor(95% to 100% International Annealed Copper Standard (IACS) conductivity.

$$I = C \times 10^6 A \sqrt{\frac{1}{t} \log_{10} \frac{T_f - 20 + (25400/G)}{T_i - 20 + (25400/G)}} \qquad \text{(Eq. 09.2)}$$

C = 3.410×10^{-4}

The variables are as defined above.

4.0 NON-RIGID OR STRAIN BUS

A strain bus is ideally suited for connecting the transmission line conductor to and from a substation bay, thus acting as a "through connection". The flexible conductors used for strain bus construction must be carefully positioned to prevent contact with other equipment and infringement upon minimum electrical clearances as specified in TES-P-119.08 under all loading conditions.

The following points shall be considered in designing a strain bus:

4.1 Conductor Size

The normal type and size of conductors to be used shall be the largest of the line conductors terminating at the substation. These shall have adequate ampacity based on the factors indicated in Clause 3.0 above, and also be of sufficient size to withstand operational requirements and system faults as specified in TES-P-119.02 without damage from overheating.

4.2 Conductor Sag and Tension

The conductor sag and tension applied to the conductor, strain insulator and related hardware shall be within the safe limits specified in SEC Engineering Standard number TES-P-122.03.

The conductor sag shall be such that the deflection during short circuit condition at maximum temperatures will provide at least the minimum phase-to-phase clearances specified in TES-P-119.08. Short circuit current used in calculating the electrical and mechanical forces shall be as specified in TES-P-119.02.

5.0 RIGID BUS

Rigid bus arrangement is most commonly used in low profile substations.

Unless otherwise specified, aluminum alloy tubing (schedule 40/80), 6101 temper T6 per ASTM B 317 shall be used for rigid buses. Accordingly, in the equations indicated in this standard the modulus of elasticity for aluminum alloy tubing where applicable is to be taken as 6.895×10^7 kPa.

In designing a rigid bus, the following aspects shall be considered in addition to the factors indicated in Clause 3.0 above:

5.1 Conductor Size

The bus tubing shall be of sufficient size and ampacity to withstand operational conditions and system faults as specified in 01-TMSS-01 and TES-P-119.02 without damage due to overheating.

5.2 Conductor Vibration

Conductor vibration is induced either due to alternating current or laminar wind flow. These frequencies are near the natural frequencies of the bus-conductors.

5.2.1 Natural Frequency (f_b) of the conductor span is given by:

$$f_b = \frac{\pi K^2}{2L^2}\sqrt{\frac{JE}{m}} \qquad \text{(Eq. 09-3)}$$

Where

f_b	=	Condcutor natural frequency (Hz)
K	=	1.00 for two pinned (simple support) ends (dimensionless)
	=	1.25 for one pinned and one fixed end (dimensionless)
	=	1.51 for two fixed ends (dimensionless).
L	=	span length, m
J	=	moment of inertia of cross sectional area, m^4 (Eq.09-13)
E	=	modulus of elasticity, N/m^2.
m	=	mass per unit length, kg/m

Table 09-1: Modulus of elasticity E for common conductor alloys

Bus-Conductor Alloy	*E*	
	kPa	lbf/in2
Aluminum 6061-T6	6.895×10^7	10×10^6
Copper	11.03×10^7	16×10^6

5.2.2 Current Induced Vibration

When current flows, magnetic fields interact and exert forces in parallel conductors. The calculated natural frequency (f_b) of the bus span as determined by Equation 09-3 shall not be more than the rated power frequency i.e., 60Hz for SEC System.

5.2.3 Wind Induced Vibration

Long conductor spans can be damaged due to aeolian vibration caused by steady low velocity transverse winds. The maximum frequency of the aeolian force for circular conductor is calculated from Strovhal formula:

$$f_a = \frac{CV}{d} \qquad \text{(Eq. 09-4)}$$

Where

f_a	=	maximum aeolian frequency, Hz
V	=	Wind velocity (m/s)
C	=	Strouhal number and approximately equal to 0.19 for a cylinder shape (dimensionless)
D	=	diameter of conductor, m

5.2.4 Vibration Attenuation

5.2.4.1 Damping may be accomplished by installing stranded bare conductor inside the tubular bus to dissipate vibrational energy. The damping conductor shall be of the same material as the tubular bus to prevent corrosion and shall be free from bends to prevent them from jamming against the tubes. The weight of the damping conductor should be from 10% to 33% of the bus conductor weight. Maximum vibration free span length for Schedule 40 tubing shall be per Table 19 of IEEE 605 (2008).

5.2.4.2 Another method used to dampen vibration consists of installing internal or external commercially available prefabricated bus dampers on the bus conductors. Size, location and installation shall be in accordance with manufacturer's instructions.

5.3 Wind Forces

Factors that affect wind forces are the speed and gust of the wind, radial ice thickness, shape, diameter, height and exposure of the conductors.

The wind force per unit length of the conductor is given by:

$$F_W = C\, C_D K_Z G_F V^2\, I(d+2r_i) \qquad \text{(Eq. 09-5)}$$

Where

F_W = wind force per unit length of bus, N/m (Ibf/ft)

C = 0.613 [2.132×10^{-4} for English units]

C_D = Drag coefficient

The drag coefficient for smooth tubular conductors is 1.0. For other shapes, are given in table 09-3.

K_Z = Height and exposure factor

ASCE 7-05 has detailed definitions of surface roughness and exposure categories which are summarised as:

- Surface roughness B: Urban or suburban areas with closely spaced obstructions such as wooded areas, single-family homes or large structures.
- Surface roughness C: Open terrain with scattered obstructions having height generally less than 9.1m (30 ft) including open country, grasslands, and water surfaces in hurricane-prone regions.

- Surface roughness D: Flat, unobstructed areas and water surfaces outside hurricane-prone regions including smooth mud flats, slat flats and unbroken ice.
- Exposure B: shall apply where the ground conditions as described by surface roughness B exist prevailing in the upwind direction a distance of atleat 800m (2630 ft).
- Exposure C shall apply for all cases where Exposure B or D do not apply.

  Exposure D: shall apply where the ground conditions as described by surface roughness D exist prevailing in the upwind direction a distance of at leat 1525 m (5000 ft). Exposure D shall extend inland a distance of 200m (660 ft) or 10 times height of the structure whichever is greater.

  Refer Table 9.02 for K_z values

Gf = Gust response factor of 0.85 shall be used on rigid bus conductors used in rigid bus structures for all exposure categories.

V = Wind speed with ice, m/s

d = outside conductor dia, m

r_i = radial ice thickness, m

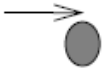
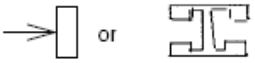
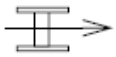
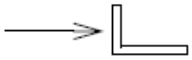
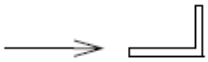
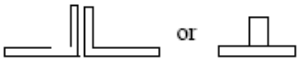
I = Importance factor for substation to be taken as 1.15 as classified by ASCE 7-02.

For indoor substations the value of $F_W = 0$

Table 09-2 : Height and exposure factor K_z

Height above ground (m)	Exposure B	Exposure C	Exposure D
0-4.6	0.57	0.85	1.03
6.1	0.62	0.90	1.08
7.6	0.66	0.94	1.12
9.1	0.70	0.98	1.16
12.2	0.76	1.04	1.22
15.2	0.81	1.09	1.27
18	0.85	1.13	1.31
21.3	0.89	1.17	1.34
24.4	0.93	1.21	1.38
27.4	0.96	1.24	1.40
30.5	0.99	1.26	1.43

Table 09-3: Drag coefficients for structural shapes

Profile and Wind Direction	C_D
	2.03
	1.00
	2.00
	2.04
	2.00
	1.83
	1.99

5.4 Short Circuit Current Forces

The magnetic fields produced by short circuit current cause forces on the bus conductors. The bus conductors and bus supports must be strong enough to withstand these forces.

The force imparted to the bus structure by short circuit current is dependent on conductor spacing, magnitude of fault current, type of short circuit, and degree of short-circuit asymmetry. Other factors to be considered are support flexibility, and corner and end effects.

Basic force equation between infintely long conducotrs is as follows:

$$F_{sc} = \frac{16\Gamma Isc^2}{10^7 D} \text{ (Metric Unit) (Eq. 9.06)}$$

Where F_{sc} = Short circuit current force by unit length , N/m
I_{sc} = symmetrical short-circuit current, A, rms
D = conductor spacing center-to-center, m
Γ = the constant based on type of short circuit and conductor location (See Table 09-4).

Equation 09-6 is very conservative and not realistic in practice and the following corrected basic force equation is proposed for infinitely long conductors.

$$F_{sc\ corrected} = D_f^2\ K_f . F_{sc} \qquad \text{(Eq. 09-7)}$$

K_f = monunting-structure flexibility factor see (Figure 09-02)

D_f = decrement factor is given by the following equation:

Where

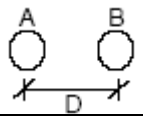
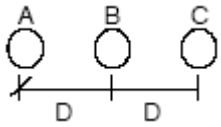
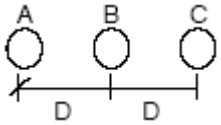
$$D_f = \frac{1 + e^{-\frac{1}{2fT_a}}}{2} \qquad \text{(Eq. 09-8)}$$

Where

$$T_a = \frac{X}{R}\frac{1}{2\pi f} \qquad \text{(Eq. 09-9)}$$

Where X=System Reactance, R=System Resistance, and f =60Hz

Table 09-4: Constant Γ for calculating short circuit current forces

Type of short circuit	Configuration	Force on conductor	Γ
Phase to Phase		A or B	1.00
Three phase		B	0.866
Three phase		A or C	0.808
Phase to phase	Triangular arrangement – equilateral triangle-side D	A or B	1.0
Three phase	Triangular arrangement – equilateral triangle-side D	A or B or C	.05

5.5 Corner and end effects

The value for the short-circuit current force calculated by equations (Eq. 09-7) are for parallel and infinitely long conductors. The results for short bus lengths will be conservative because of end effects. The equations cannot be used for special cases, such as corners and nonparallel conductors.

5.6 Gravitational Forces

In some rigid-bus structure configurations, the insulator may be subjected to cantilever gravitational forces. These forces should be added vectorially to the fault current and wind forces. These gravitational forces will be due to the mass of the supported rigid bus, the mass of the insulator itself or other concentrated masses, or both. The effective weight of the bus mass transmitted to the bus-support fitting can be determined using Equation (Eq. 09-10).

$$F_{GB} = L_E F_G \qquad \text{(Eq. 09-10)}$$

Where F_{GB} = effective weight of bus transmitted to bus-support fitting, N [lbf]

L_E = effective bus span length, m [ft] (see Table 09-6)

F_G = total bus unit weight, N/m [lbf/ft] (see Eq. 09-11)

If the bus span is subjected to concentrated loads, the force transmitted to the bus-support fitting should be analyzed more thoroughly.

The weight of the insulator F_{GI} should be included in the total cantilever force if the insulator is not mounted vertically.

5.7 Vertical Deflection

The total gravitational force on a conductor is the sum of the weights of the conductor, ice and damping material,and any concentrated loads. Without concentrated loads,

$$F_G = F_C + F_I + F_D \qquad \text{(Eq. 09-10)}$$

Where
F_G = total bus unit weight, N/m [lbf/ft]
F_c = conductor unit weight, N/m [lbf/ft]
F_I = ice unit weight, N/m [lbf/ft]
F_D = damping material unit weight, N/m [lbf/ft]

If the bus span is subjected to concentrated loads, the force distribution on that span should be analyzed more thoroughly.

5.7.1 Vertical Deflection Limits

The vertical deflection shall not exceed either 1 in 200 of span length or 0.5 to 1.0 times the vertical dimension (diameter, for cylindrical conductor) of the bus conductor.

5.7.2 Allowable Span Length for Vertical Deflection Considerations

The maximum allowable bus span length for a given deflection and vertical loads for different end conditions may be calculated from either of the Equations given below:

<u>Pinned-Pinned condition</u>

$$L_D = \left[\frac{384EJ\,\delta_{\max}}{5F_G}\right]^{\frac{1}{4}} \qquad \text{(Eq. 09-11)}$$

Where
L_D =Allowable span on the vertical deflection limit, m [in]
E =Young's modulous of conductor material, N/m[Ibf/in^2]
J = Bending moment of intertia of the conductor cross section m^4 [in]
F_G = Gravitational weight by unit length N/m [lbf/in]
δ_{max} =Vertical deflection limit, m [in]

$$L_D = \left[\frac{384EJ\eta}{5F_G}\right]^{\frac{1}{3}} \qquad \text{(Eq. 09-12)}$$

where η is the fraction of allowable span

Bending movement J of a circular cross section is given by

$$J = \frac{(D_o^4 - D_i^4)}{64} \qquad \text{(Eq. 09-13)}$$

Where D_o = Conductor outside diameter, m [in]
D_i = Conductor inside diameter, m [in]

<u>Fixed-Fixed end condtition</u>

$$L_D = \left[\frac{384EJ\delta_{\max}}{F_G}\right]^{\frac{1}{4}} \text{ and } \left[\frac{384EJ\eta}{F_G}\right]^{\frac{1}{4}} \qquad \text{(Eq. 09-14)}$$

<u>Pinned-Fixed end conditions</u>

$$L_D = \left[\frac{185EJ\delta_{\max}}{F_G}\right]^{\frac{1}{4}} \text{ and } \left[\frac{185EJ\eta}{F_G}\right]^{\frac{1}{4}} \qquad \text{(Eq. 09-15)}$$

5.8 Fiber Stress Considerations

In some cases, span length may be limited by the fiber stress of the bus conductor material. In practice, when wind and gravitational forces are combined, the elastic limit stress is commonly used as the maximum allowable stress. When wind, gravitational and short circuit current forces F_{SC} are combined, the minimum yield stress is commonly used as the maximum allowable stress, since F_{SC} is conservative.

The total force on a conductor in horizontal configuration is given by:

$$F_T = \sqrt{(F_W + F_{SC})^2 + (F_G)^2} \qquad \text{(Eq. 09-16)}$$

Where
F_T = Resultant unit Force on conductor, N/m [Ibf/in]
F_W = Unit Wind Force, N/m[Ibf/in] (refer Eq. 09-5)
F_{SC} = Unit Short circuit Force, N/m , [Ibf/in] (refer Eq. 09-6)
F_G = Total bus unit weight inclusive of superimposed weights, if any, N/m, [Ibf/ft]

5.8.1 Maximum allowable span length from fiber stress consideration

Single-span bus, pinned-pinned end conditions

$$L_s = \sqrt{\frac{16J\sigma_{allowable}}{F_T D_o}} \text{、} \qquad \text{(Eq. 09-17)}$$

Where

L_S = Maximum allowable span length, m [in]
J = Bending moment of intertiaof the conductor cross section m^4 [in^4]
F_T = Total force acting on the conductor by unit length N/m [lbf/in]
$\sigma_{allowable}$ Allowable stress of the conductor material, N/m2 [lbf/in2]
D_o = Conductor outside diameter, m [in]

Single-span bus, fixed-fixed end condtions

$$L_s = \sqrt{\frac{24J\sigma_{allowable}}{F_T D_o}} \text{、} \qquad \text{(Eq. 09-18)}$$

Single-span bus, pinned-fixed end condtions

$$L_s = \sqrt{\frac{16J\sigma_{allowable}}{F_T D_o}} \text{、} \qquad \text{(Eq. 09-19)}$$

5.8.2 For a continuois bus, the evaluation of allowable span length based on fiber stress is more compled and be analyzed more thoroughly.

6.0 INSULATOR STRENGTH CONSIDERATIONS

In addition to the requirements of TES-P-119.11, the insulators shall be capable of withstanding the total bus loading and shall have sufficient electromechanical strength to withstand the short circuit forces.

A bus support insulator has to withstand forces resulting from wind, short circuit and gravity. These forces basically cause cantilever effect on the insulator. This effect shall not exceed the cantilever strength of the insulator. In addition, the tension, torsion and compression forces, as applicable, on the insulator shall also be considered.

6.1 Effective Bus Span Length

Effective bus span length L_E is used for calculating wind, gravitational and short circuit forces on bus conductor. The values of L_E for various end conditions are given in Table 09-6.

If end conditions are not known, then the condition which yields the maximum L_E shall be used.

Table 09-6: Effective Bus Span Length L_E Supported by Insulator

Bus Configuration	Support Conditions					Maximum Span Length L_E
	S1	S2	S3	S4	S5	
Single Span	Pinned	Pinned				1/2×L
Single Span	Pinned	Fixed				5/8×L (max at S2)
Single Span	Fixed	Fixed				1/2× L
Two Continuous Span	Pinned	Continuous	Pinned			5/4×L (Max at S2)
Two Continuous Span	Pinned	Fixed	Fixed			9/8×L (max at S2)
Two Continuous Span	Fixed	Fixed	Fixed			L(Max at S2)
Three Continuous Span	Pinned	Continuous	Continuous	Pinned		11/10×L (max at S2)
Three Continuous Span	Pinned	Continuous	Continuous	Continuous	Pinned	32/28×L (max at S2)

Where L = Bus Span length-Equal Spans & Continuous: means Mid-Support of Continuous Span

6.2 Bus Short Circuit Current Force

This bus short circuit current force transmitted to the bus support fittings can be calculated from the Equation

$$F_{SB} = L_E F_{SC} \qquad \text{(Eq. 09-20)}$$

Where F_{SB} = Bus short circuit current force transmitted to the bus support fittings, N [Ibf]

L_E = Effective span length, m [ft], (See Table 09-6)

F_{SC} = Short circuit current unit force, N/m [Ibf/ft] (as calculated in clause 5.4-Eq. 09-6)

6.3 Bus Wind Force

The bus wind force transmitted to the bus support fittings is given by:

$$F_{WB} = L_E F_W \qquad \text{(Eq. 09-21)}$$

Where F_{WB} = Bus Wind Force transmitted to the bus support fittings, N [Ibf]
L_E = Effective span length, m [ft], (See Table 09-6)
F_W = Wind unit force on bus, N/m [lbf/ft] (refer clause 5.3-Eq. 09-5)

6.4 Insulator Wind Force

The insulator wind force acting at the center of an insulator is given by:

$$F_{WI} = C\, C_D\, K_Z\, G_F\, V^2\, I(D_i + 2r_i)H_i \qquad \text{(Eq. 09-22)}$$

Where F_{WI} = wind force on insulator, N [Ibf]

H_i = Insulator height, m [in]

*The effective insulator diameter D_i is to be taken as diameter of insulator at skirt. If the insulator consists of stacks of different diameters, the effective diameter D_i is given by

$$D_i = \frac{D_1 + D_2 + \cdots D_n}{n} \qquad \text{(Eq. 09-23)}$$

Where D_1, D_2, ...D_n are diameters at skirts of stack 1, 2, ...n

For indoor substations, the value of F_{WI}= 0.

6.5 Gravitational Forces

The effective weight of the bus mass transmitted to the bus support fitting may be determined by:

$$F_{GB} = L_E F_G \qquad \text{(Eq. 09-24)}$$

Where F_{GB} = Effective weight of bus transmitted to bus support N, [Ibf]
L_E = Effective span length in meters, [ft] (See Table 09-6)
F_G = Total bus unit weight inclusive of superimposed weights, if any, N/m [Ibf/ft]

If the insulator is not mounted vertically, its weight F_{GI} shall be included in the total cantilever force.

6.6 Total Insulator Cantilever Load

6.6.1 Vetically Mounted Insulator

Load acting on the insulator comprises of load on the horizontal direction perpendicular to bus (x direction) load in the bus direction (y direction) and vertical direction (z direction).

Equivalent cantilever force in x direction due to F_{SB}, F_{WI} and F_{WB} is given by the following equlation:

$$F_{c\text{-}x} = \frac{F_{WI}}{2} + \frac{(H_i + H_f)F_{WB}}{H_i} + \frac{(H_i + H_f)F_{SB}}{H_i} \qquad \text{(Eq. 09-25)}$$

Where H_f = Height from insulator top to bus center line, m

Cantilever force in y direction:

$$F_{c\text{-}y} = F_{TE}\frac{(H_i + H_f)}{H_i} \qquad \text{(Eq. 09-26)}$$

Where F_{TE} = Thermal force generated in the conductor due to expansion

F_{TE} = $E\ A\ \alpha\ (T_f - T_i)$

Where F_{TE} = Thermal force, N [lbf]
E = Conductor material modulous of elasticity, N/m^2 [lbf/in^2]
A = Conductor cross section area, m^2 [in^2]

α = Coefficient of thermal expansion (1/°C)

Total cantilever load on an insulator is the summation of the cantilever forces acting on the insulator, multiplied by their overload factors.

Total cantilever strength due to application of load in x and y direction:

$$F_C \quad \sqrt{(F_{c\text{-}x}^2 + F_{c\text{-}y}^2)} \qquad \text{(Eq. 09-27)}$$

6.6.2 Horizontally Mounted Insulator

Total cantilever force due to various force are given by the following equation:

$$F_C = \sqrt{\left(\frac{F_{GI}}{2} + \frac{(H_T + H_i)(F_{GB} + F_{SB} + F_{CCi})}{H_T}\right)^2 + \left(\frac{F_{WI}}{2} + F_{TE}\left(\frac{H_T + H_f}{H_T}\right)\right)^2} \quad \text{(Eq. 09-28)}$$

Where F_{WI} and F_{TE} are acting in y direction and F_{SB}, F_{GB}, F_{CCi} (concentrated load), F_{GI} (cantilever force from insulator weight) in z direction.

7.0 INSULATORS SUPPORTING TWO LEVEL BUS CONDUCTORS

The wind forces and short circuit forces on the high and low buses are determined individually based on the method and formulae given in this standard.

7.1 Low Bus

For low bus, these may be denoted as:

F_{SBL} = Short circuit current force, N [Ibf] (refer Eq. 09-20)
F_{WBL} = Wind force, N[Ibf] (refer Eq. 09-21)

7.2 High Bus

For high bus, these may be denoted as:

F_{SBH} = Short circuit current force, N [Ibf] (refer Eq. 09-20)
F_{WBH} = Wind force, N [Ibf] (refer Eq. 09-21)

7.3 Low Bus to Insulator Top

Factor for transfer of force from low bus to insulator top is determined by

$$F_{LBI} = \frac{H_i + H_f}{H_i} \quad \text{(Eq. 09-29)}$$

Where F_{LBI} = Factor for transfer of force
H_i and H_f as in Eq. 09-23

7.4 High Bus to Insulator Top

Factor for transfer of force from high bus to insulator top is determined by

$$F_{HBI} = \frac{H_A + H_f + H_i}{H_i} \quad \text{(Eq. 09-30)}$$

Where F_{HBI} = Factor for transfer of force
H_A = A frame height, cm (refer Figure 09-3)
H_i and H_f as in Eq. 09-23

7.5 A-Frame to Insulator Top

Factor for transfer of force from A-frame to insulator top is determined by

$$F_{AI} = \frac{0.5H_A + H_f + H_i}{H_i} \qquad \text{(Eq. 09-31)}$$

Where F_{AI} = Factor for transfer of force
H_A, H_f and H_i as in Eq. 09-23 and Eq. 09-26

7.6 Wind Force on A-Frame

The wind force on the A-Frame is given by:

$$F_{WA} = F_W.(2H_A) \qquad \text{(Eq. 09-32)}$$

when only height of A-frame is exposed to the direction of wind, usually when direction of wind is perpendicular to high bus

$$F_{WA} = F_W.2H_A(\sec \theta/2 + 0.5 \tan \theta/2) \qquad \text{(Eq. 09-33)}$$

when full A-frame is exposed to the direction of wind flow, usually when wind is perpendicular to low bus.

Where F_{WA} = Wind Force on A-frame, N
F_W = Unit wind Force on bus, N/m (refer Eq.09-5)
(here FW may be calculated for low bus)
H_A = Height of A - frame, cm (refer Figure 09-3)
θ = angle of opening of A-frame.

7.7 Total Cantilever Load on Insulator

Total vectorial sum of forces on insulator with wind perpendicular to high bus and short circuit forces on both high and low buses is given by:

$$F_{SH} = K_1\left[\frac{F_{WI}}{2} + F_{HBI} \cdot F_{WBH} + F_{AI} \cdot F_{WA}\right] + K_2\left[F_{HBI} \cdot F_{SBH} + F_{LBI} \cdot F_{SBL}\right] \qquad \text{(Eq. 09-34)}$$

Total vectorial sum of forces on insulator with wind perpendicular to low bus and short circuit forces on both low and high buses is given by

$$F_{SL} = K_1\left[\frac{F_{WI}}{2} + F_{LBI} \cdot F_{WBL} + F_{AI} \cdot F_{WA}\right] + K_2\left[F_{LBI} \cdot F_{SBL} + F_{HBI} \cdot F_{SBH}\right] \qquad \text{(Eq. 09-35)}$$

Where F_{SH} = Total cantilever load acting at the top of the insulator when wind is perpendicular to high bus, N[Ibf]

F_{SL} = Total cantilever load acting at the top of the insulator when wind is perpendicular to low bus, N [Ibf]

F_{WI} = wind force on insulator, N [Ibf]

F_{HBI} = Factor for transfer of force, high bus (refer Eq. 09-30)

F_{LBI} = Factor for transfer of force, low bus (refer Eq. 09-29)

F_{WBH} = wind force for high bus, N [Ibf] (refer Clause 7.2)

F_{WBL} = wind force for low bus, N [Ibf] (refer Clause 7.1)

F_{AI} = Factor for transfer of force from A-frame (refer Eq. 09-31)

F_{WA} = Wind force on A-Frame, (refer Clause 7.6)

F_{SBH} = Short circuit current force for high bus, N [Ibf] (refer Clause 7.2)

F_{SBL} = Short circuit current force for low bus, N [Ibf] (refer Clause 7.1)

K_1 = Overload Factor applied to wind forces, may be taken as 2.5

K_2 = Overload factor applied to short circuit current forces, as specified in clause 6.6.2.

The above mentioned cantilever forces are to be taken into consideration in fixing the insulator strength.

7.7.1 If the A-frame strength is found inadequate to withstand the forces applicable, suitable double insulator supports at A-frame locations may be considered.

8.0 CORONA & CONDUCTOR SURFACE VOLTAGE GRADIENT

Corona on high voltage substations occurs on conductors, hardware and insulators as a field sustained electrical breakdown of air which produces visible light, audible noise, energy loss, radio interference, chemical action and mechanical vibration. The occurrence of corona is determined by the conductor surface voltage gradient.

The allowable surface voltage gradient is given by:

$$E_o = \delta g_o \qquad \text{(Eq. 09-36)}$$

Where

E_o = allowable surface voltage gradient kVrms/cm

g_o = allowable voltage gradient under standard conditions for equal radio influence generation for circular conductors, kVrms/cm (See Figure 09-2)

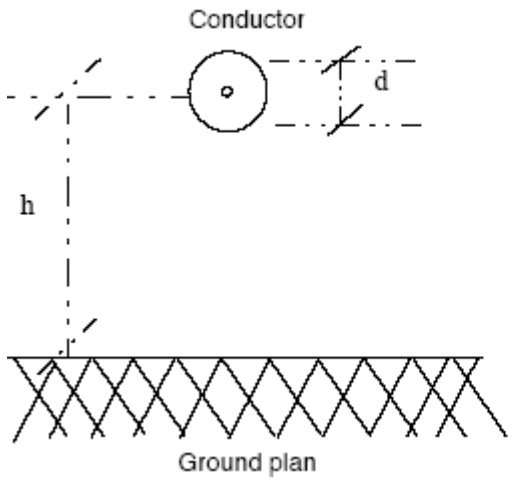
δ is air density factor defined as:

$$\delta = \frac{3.9211\,b}{273 + t} \qquad \text{(Eq. 09-37)}$$

where b = barometric pressure, cm of Hg
t = temperature, ºC

The maximum surface voltage gradient E_m at the surface of smooth circular conductors, at operating voltage may be determined as follow:

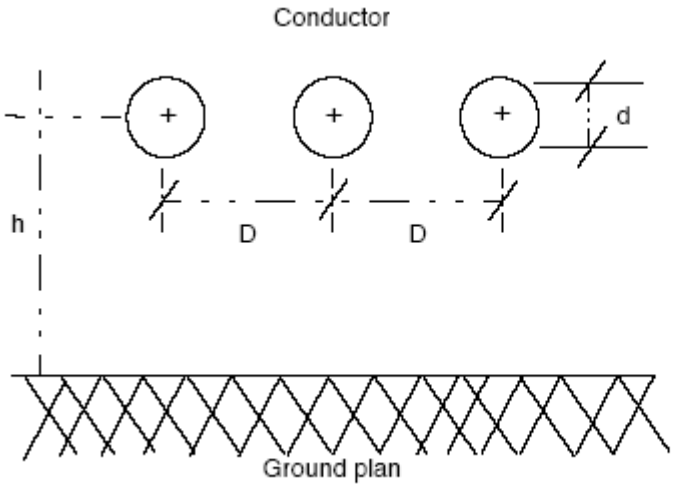
<u>For single phase conductor</u>:



$$E_a = \frac{V_1}{\frac{d}{2} L_n\left(\frac{4h}{d}\right)}$$

$$E_m = \frac{h}{h - \frac{d}{2}} E_a$$

For three phase conductors:



$$E_a = \frac{V_1}{\frac{d}{2} L_n\left(\frac{4h_e}{d}\right)}$$

$$E_m = \frac{h_e}{h_e - \frac{d}{2}} E_a$$

$$h_e = \frac{hD}{\sqrt{4h^2 + D^2}}$$

Where

E_m	=	maximum voltage gradient at the surface of the conductor, kV/cm
E_a	=	average voltage gradient at the surface of the conductor, kV/cm
h	=	distance from center of the conductor to ground plane, cm
h_e	=	equivalent distance from the center of the conductor to ground plane for three phase, cm
d	=	diameter of the individual conductor, cm
D	=	phase to phase spacing for three phase, cm
V_1	=	line to ground test voltage.kV which is taken as 110% of nominal operating line to ground voltage

For satisfactory operation, E_m must be less than E_o.

At extra high voltage commercially available corona rings shall be used to contain the surface voltage gradient.

9.0 BUS SUPPORT STRUCTURES

9.1 Rigid Conductor Support Structures

Bus support structures shall be constructed from steel pipe, I-beam or square tube sections conforming to ASTM A53 Grade B and ASTM A501 respectively. Steel shall be galvanized in accordance with ASTM A123, with the exception that the minimum coating thickness shall be as per 01-TMSS-01.

9.2 Flexible Conductor Support Structures

These structures commonly known as gantry/take-off structures are used for termination of the slack spans from dead-end or terminal towers on one side and flexible buses on the other side. In many cases these structures may only be used for

termination of slack spans from dead-end towers or only flexible strung buses across the substation bays. These structures are therefore to be designed for one side tension/pulls only.

The main forces acting on these structures are the tensions of overhead ground wires and conductors, wind and gravitational forces. It is for the designer to check all possible combinations of loads/forces for determining stresses in various elements of gantry structures.

Basically there are two main components of gantry structures, i.e. columns and beams. These can be fabricated from angle sections or other steel profiles, the selection of which shall be based on economical considerations.

Material for lattice type columns and beams shall conform to 20-TMSS-01 and shall be hot dip galvanized with the thickness of coating as specified therein. All connections shall be of bolted type.

10.0 EQUIPMENT SUPPORT STRUCTURES

These structures hold the substation equipment in position, with sufficient rigidity to assure the satisfactory functions of the equipment. These shall be of lattice steel type construction and shall be designed to withstand apparatus loads, weight of structure, line tension, wind loads, snow/ice loads if appliacble, magnetic forces due to short circuit currents, operating and dynamic loads due to operation of apparatus such as air switches, ground switches and interrupting devices as specified by the manufacturer of the equipment.

In designing equipment supports, no consideration will be given to restraint that may be offered by the various connections of the equipment. The supports shall be capable of standing free. These structures shall be designed in accordance with ASCE Standard 10-97.

11.0 TAKE-OFF STRUCTURES/ GANTRY

Substation structures shall be designed for the loads that are listed below:

Take-off structure/gantry structure and strain bus termination structures shall be designed to anchor slack tensions and/or bus tensions, to support their own weight, the weight of the anchored cables and the weight and operating loads of any equipment installed thereon, as well as to withstand wind load from any direction acting on the structure, conductors, equipment and the short circuit forces.

11.1 Effective Spans

Effective spans for the purpose of load calculations shall be as under:

a. Wind span = ½ sum of slack span and bus span
b. Weight span = ½ sum of slack span and bus span

Slack spans are normally depends on the OHTL layout and location of Take-off structure/gantry structure

11.2 Types of Loads

11.2.1 Horizontal Loads

a. Horizontal loads on beams are due to:

i. Tensions of bus conductors/ slack span tensions.
ii. Wind loads on bus conductors, slack spans based on effective wind span.
iii. Wind loads on insulator strings and fittings.
iv. Wind loads on beam members.
v. Short circuit current loads.

b. Horizontal loads on columns are due to:

i. Tensions of ground wire(s).
ii. Wind loads on ground wire(s) based on effective wind spans.
iii. Wind on ground wire fittings.
iv. Reaction of beam at the point of connections with columns.
v. Wind loads on column members.

11.2.2 Vertical Loads

a. Vertical loads on beams are due to:

i. Weight of bus conductors and slack spans based on effective weight span.
ii. Weight of insulators, fittings, any equipment mounted on the beam.
iii. Weight of lineman and tools, to be assumed equal to 150kg f or 1.5 kN.
iv. Self weight of beam.

b. Vertical loads on columns are due to:

i. Weight of ground wire(s) based on effective weight span.
ii. Weight of lineman and tools (150kg for 1.5kN).
iii. Vertical reaction of beams at the point of connection.
iv. Self weight of columns.

11.3 Loading

11.3.1 Tensions

Tensions in slack spans and bus conductors shall be kept at a level which shall not cause heavy loading and overturning moments on gantry structures and, at the same time, do not result in excessive sags in slack spans. An optimum value of slack tension is around 30% of maximum final line tension.

Slack span shall be considered normal (perpendicular) to the longitudinal axis of the beam. However, design of gantry structures shall be suitable for deviation angle of slack span up to 30 degrees from normal.

11.3.2 Wind Pressure

Wind pressure on conductor and ground wire on full projected area = 1.064 kPa

Wind pressure on insulator strings and fittings = 1.064 kPa

Wind pressure on structures on 2.0 times the projected area of members on windward face = 1.6 x 1.064 kPa

11.3.3 Short Circuit Current Forces

Short circuit current forces result in increase in tensions of bus conductors.The supporting structures, i.e. the columns and beams of gantry structures shall be able to withstand these additional forces. Effect and magnitude of short circuit current forces on flexible strung bus conductors shall be determined in accordance with the procedures given in IEC 60865-1.

11.3.4 Operating and Dynamic Loads

Forces resulting from operation of equipment shall also be considered in design of gantry structures if such equipment are mounted on the gantry structure.

11.3.5 Construction Load

While such conditions do not usually govern design, they should also be considered for safety of workman.

11.4 Loading combination and Factor of Safety

11.4.1 Normal condition, factor of Safety = 2.0 for

a. Loads on gantry structures due to one side tension of all conductors and ground wire.
b. Loads due to wind on conductor, ground wire(s), insulators, supporting structures.
c. Loads due to weight of conductors, ground wire(s), insulators, equipment, man and tools, self weight of structures.

11.4.2 Short Circuit Condition, Factor of Safety = 1.5 for

a. Loads on gantry structures due to one side tension of all conductors and ground wire(s).
b. Loads due to wind on conductor, ground wire(s), insulators, supporting structures.
c. Loads due to weight of conductors, ground wire(s), insulators,equipment, self weight of structure.
d. Loads due to short circuit forces.

11.4.3 For determining wind loads on various components, only one wind direction at a time shall be considered.

12.0 STRESSES

12.1 Axial Stresses in Tension

The estimated tensile stresses on the net effective sectional area in various members, multiplied by the appropriate factor of safety shall not exceed minimum guaranteed yield stress of the material.

12.2 Axial Stresses in Compression

The estimated compressive stresses in various members multiplied by the appropriate factor of safety shall not exceed the stress value obtained from strut formulae given in ASCE 10-97.

12.3 Stresses in Bolts

Stresses in bolts and all details regarding connections, joints end and edge distances shall be in accordance with ASCE Standard 10-97.

12.4 General Requirement

All other related design and fabrication details including limiting of stresses in bolts shall be in accordance with American Institute for Steel Construction (AISC) Specification for the Design, Fabrication and Erection of Structural Take-off structure/gantry structure for Buildings, if not specifically provided in ASCE Standard 10-97.

13.0 DEFLECTION CRITERIA

The deflection limitations given here are taken from the NEMA standards Publication No. SG6-2000, Part 36. These limitations apply to all equipment supports, including bus supports and strain bus anchors.

13.1 Vertical Deflection

The vertical deflection of the beams under the loadings described in clause 10.0 and 11.0 shall be limited to 1/300 of the span of the beam. Vertical deflection of the columns supporting the beams shall be ignored.

13.2 Horizontal Deflection

The total horizontal deflection of the beam plus the horizontal deflection of the supporting columns under the loading described in clause 11.0 shall be limited to 1/200 of the height between attachment point of the beam and the base of the supporting columns. Attachment point of beam may be assumed as the center line of strain plate for connecting the tension strings..

14.0 BUS CONNECTIONS

Compression type tension clamps, with barrels that are pre-filled with compound shall be used for all strain bus connections.

For rigid bus, adequate provisions shall be made for bus movement with temperature change by using slip or expansion bus supports, flexible taps and expansion terminals.

Ends of all bus runs shall be fitted with caps, preferably of the driven type. The caps shall be of sufficient strength to prevent movement of damping conductors out of the tube. For indoor systems, rigid bus supports which are only holding the bus and not solidly clamped to the bus, shall be connected by a jumper to avoid corona discharge.

Rigid aluminum bus hardware including all connections to the buses and all bus support structure hardware shall be of the welded type.

Whenever connectors are utilized for making electrical connections, they shall be electrically and mechanically compatible to the conductors themselves.

Connections to circuit breakers, power transformers, voltage transformers and other devices or terminals that could be damaged by conductor movement, shall be either with flexible conductors or expansion type connectors. Use of expansion type terminals depends on the bus configuration and other expansion points. Expansion fittings used on long horizontal buses are to be limited to those permitting horizontal expansion only.

15.0 FASTENERS

There are three types of structural bolts used in structure design:

a. ASTM A394: covers galvanized hexagonal head bolts including hexagonal nuts with sizes from 12 to 25 mm in diameter.

b. ASTM A307: covers galvanized regular square or hexagonal head bolts and nuts ranging from 6 to 100 mm in diameter. Grade A bolts shall be used.

c. ASTM A325: covers high strength bolts sizes from 12 to 40 mm in diameter. Nuts shall conform to ASTM A563 Grade DH.

Minimum spacing of bolts shall be in accordance with NEMA Publication No. SG6-2000, Section 36, Para 36.12.6 or as per ASCE 10-97. Minimum end distance will be the same as minimum edge distance.

All Bolts, nuts and washers shall be galvanized in accordance with ASTM A153.

16.0 WELDING

16.1 Structural Steel

All welding of structural steel should be in accordance with the latest edition of "Structural Welding Code," D1.1 of the American Welding Society. Structures that are to be galvanized should have, in addition to the required design welds, all joints sealed with a small continuous seal weld. This is to help prevent corrosion by ingress of acid through cracks between two pieces of abutting steel during immersion in acid bath before galvanizing. This is covered in ASTM A385.

16.2 Aluminum-to-Aluminum connections

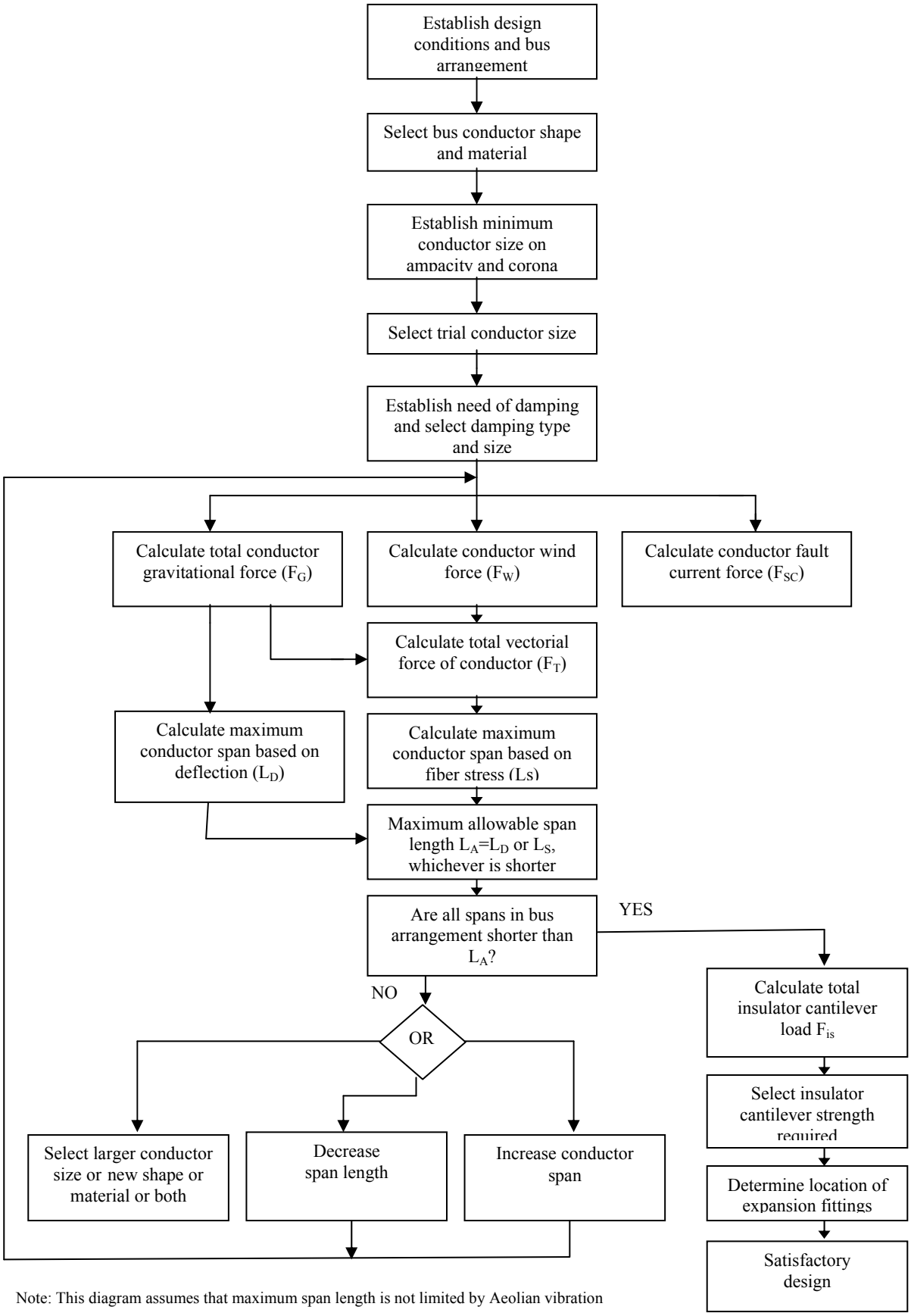
Welded Aluminum-to-Aluminum welding connections shall meet the procedures outlined in the American Welding Society Handbook "Welding Aluminum".

Details of gas tungsten-arc welding (GTAW) and gas metal-arc welding (GMAW) as to shaping of the edges to be welded, current density, size and kind of filler wire, speed of welding, etc. to be adopted are described in specialized publications and instruction manuals of aluminum producers and welding equipment manufacturers.

17.0 PHASE IDENTIFICATION TAGS

The phase identification tags (R, Y, B) shall be similar to that shown in Figure 09-4. These plates shall be installed at line terminations, at the ends of a bus run, bus tap-off positions, circuit breakers, disconnect switches and transformer positions.

In indoor substations, phase identification plates shall be at all line terminations and busbars shall be painted every 10 (ten) meters on the bus and at tap-off positions.



Note: This diagram assumes that maximum span length is not limited by Aeolian vibration

Figure 09-1: Design process for horizontal rigid bus

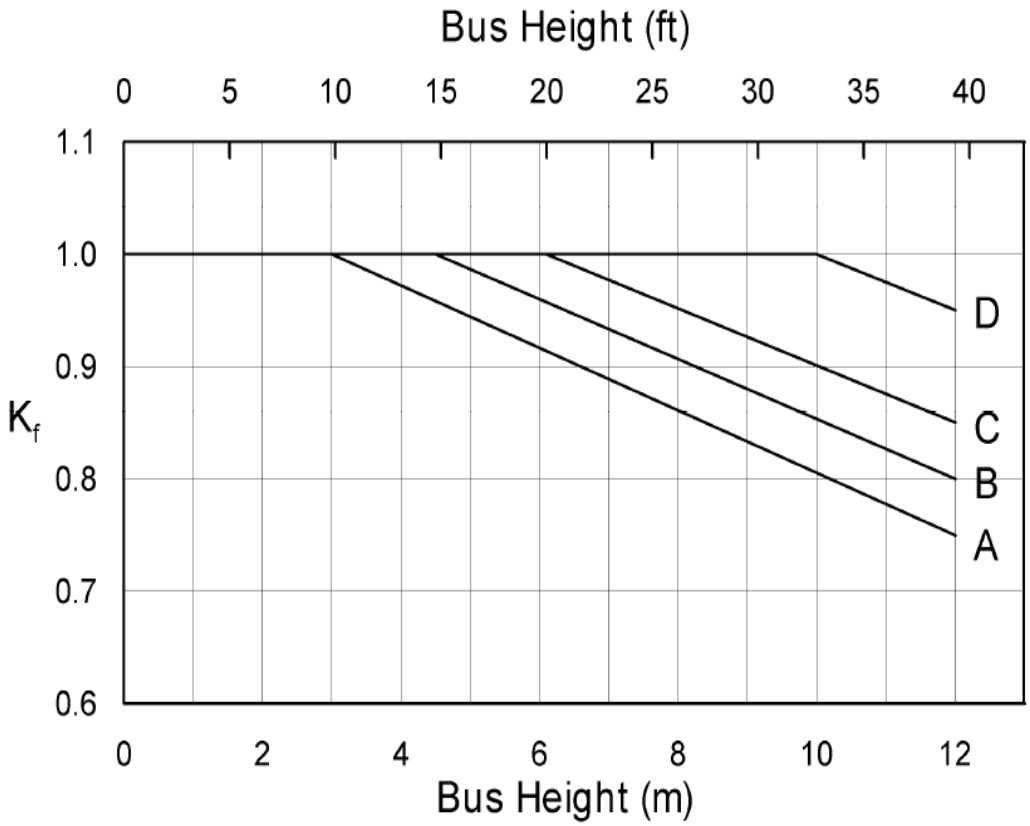


Figure 09-2: Kf Various Types of Single-Phase Mounting Structures.

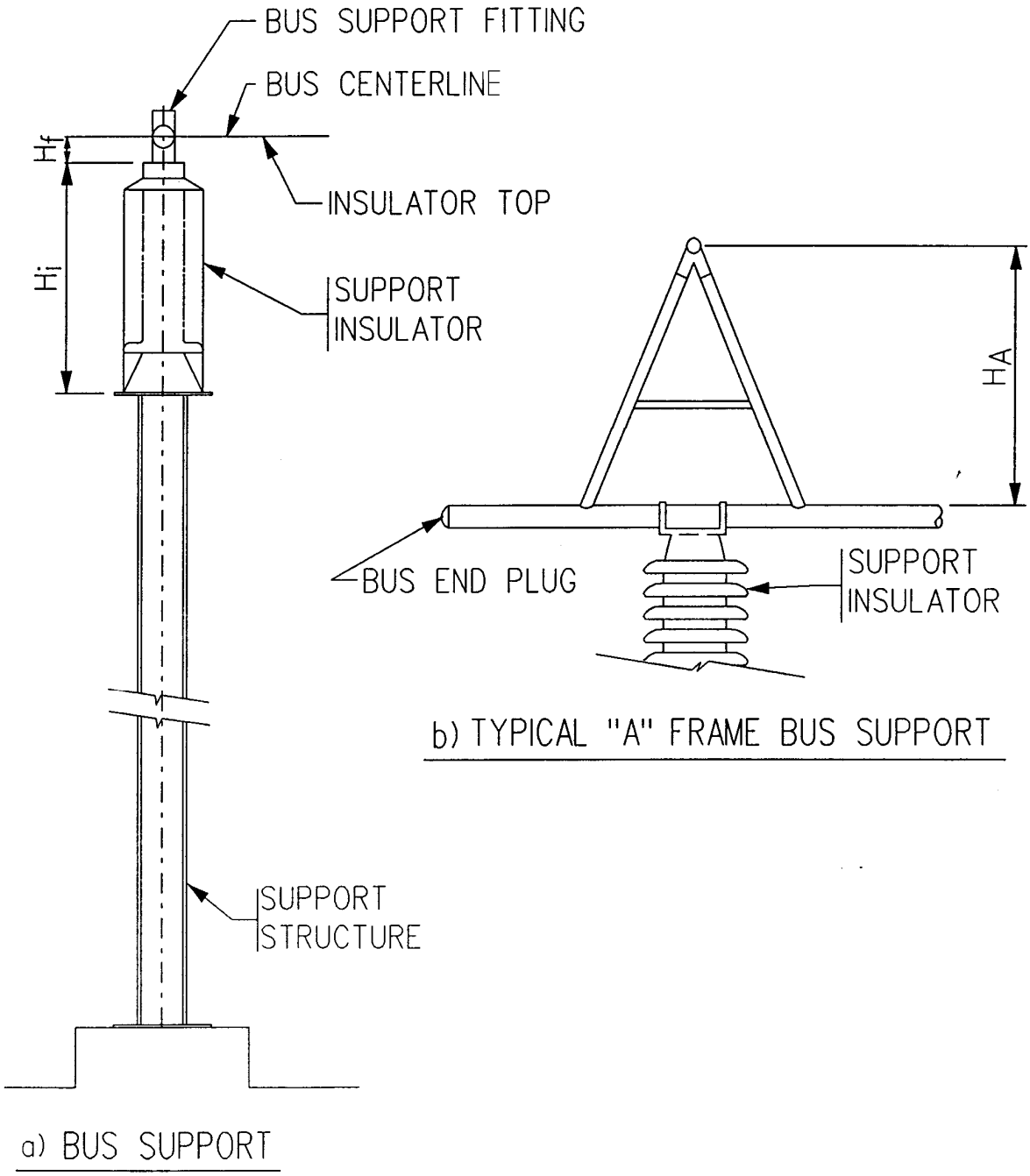
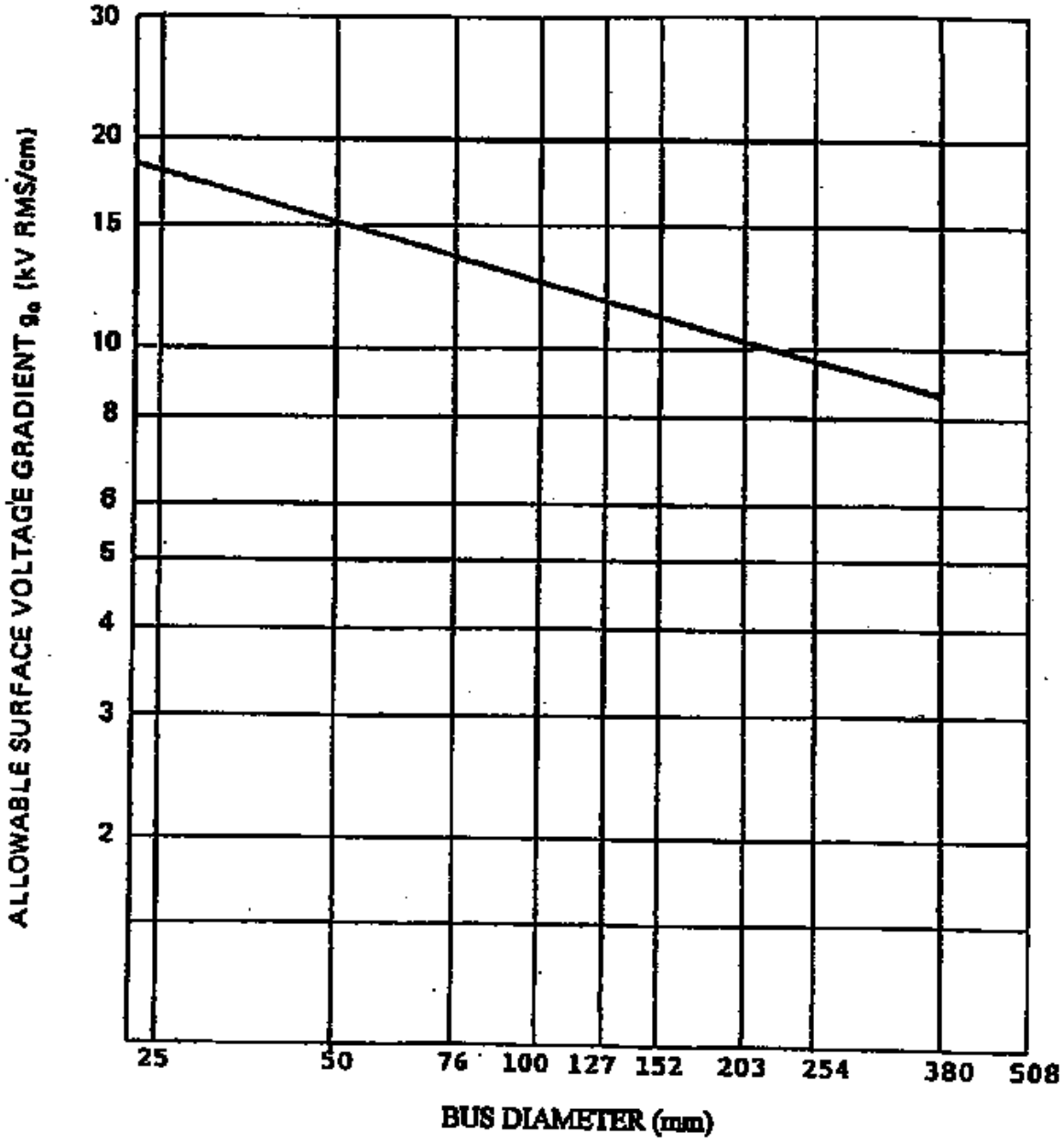
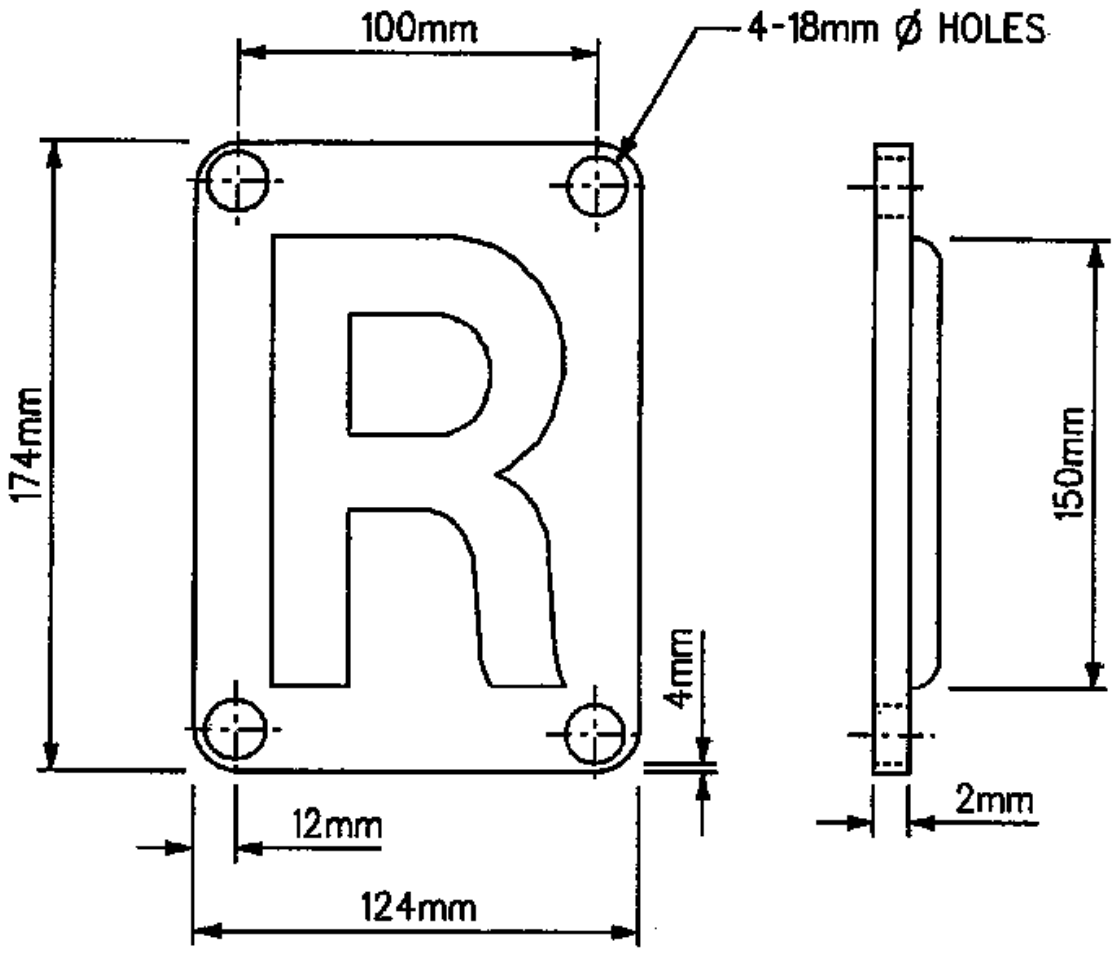


Figure 09-3: Bus Insulator Support.



Fiqure 09-04: Allowable Surface Voltage Gradient for Equal RI Generation Under Stendard Conditions Versus Bus Diameter.



Material:
Material; Andodized Aluminum

Note:
Character indentations to be filled with black baked enamel.

Figure 09-5: Phase Identification Tags

18.0 BIBLIOGRAPHY

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